

Transmission media and network devices ★ Golden

One college period · 40–45 minutes · Computer Communication & Networks

What you will learn

- Classify guided and unguided media
- Describe twisted pair, coaxial and fibre optic
- Describe radio, microwave and infrared
- Explain NIC, modem, hub, switch, router, bridge, gateway

Guided (wired) media

Medium	Construction	Use / note
Twisted pair	Pairs of copper wires twisted (UTP/STP)	LAN cables Cat5/Cat6; cheap
Coaxial	Core, insulation, metal shield, jacket	Old TV cable, some backbone
Fibre optic	Glass/plastic, light pulses	Very fast, long distance, no EMI

Unguided (wireless) media

- Radio waves: FM radio, Wi-Fi, Bluetooth, cellular.
- Microwaves: line-of-sight dishes, some mobile backhaul.
- Infrared: TV remote, old short laptop links; blocked by walls.

LAN / communication devices

Device	Job
NIC (Network Interface Card)	Fits in PC; has MAC address
Hub	Repeats bits to all ports (old, collisions)
Switch	Forwards frames to the correct port using MAC
Bridge	Connects two similar LANs / segments
Router	Connects different networks using IP; path to the internet
Gateway	Connects two different networks/protocols (e.g. LAN to mainframe)
Modem	Analog↔digital for the phone/cable line
Access point	Wi-Fi radio for wireless clients

HOW BIEK ASKS THIS

2026: coaxial and fibre are communication media; bridge connects two similar networks; router/gateway for different networks; SMTP is the common email protocol.

REMEMBER IT

Memory: Switch is smart hub. Router is smart switch that knows IP. Gateway is a protocol translator. Bridge = two similar LANs.

گائیڈڈ میڈیم تار ہے: ٹوئسٹڈ پیئر، کواکسیئل، فائبر آپٹک۔ ان گائیڈڈ ریڈیو، مائیکرو ویو، انفراریڈ۔

سوئچ LAN کے کمپیوٹر جوڑتا ہے، روٹر مختلف نیٹ ورک اور انٹرنیٹ، بریج دو ملتے جلتے LAN، گیٹ وے مختلف پروٹوکول۔

CHECK YOURSELF

1. Which cable uses light?
2. Which device uses IP addresses to connect networks?

Answers: 1. Fibre optic · 2. Router

MCQs

1. Coaxial cable and fibre optics are examples of:
 - a. Router
 - b. Communication media
 - c. Modem
 - d. Switch
2. The LAN component that connects two similar networks is:
 - a. Server
 - b. Bridge
 - c. Router
 - d. Gateway
3. The most common protocol used for e-mail is:
 - a. FTP
 - b. TCP/IP
 - c. SMTP
 - d. IEEE

Key: 1=b, 2=b, 3=c

Short questions (3 marks each)

1. Explain data communication channels / transmission media with examples.
2. Define the LAN component which allows communication between two different networks.
3. Differentiate hub and switch.

Model short answers

1. Guided: twisted pair, coaxial, fibre. Unguided: radio, microwave, infrared. Fibre is fastest and immune to electrical noise.
2. Router or gateway. A router forwards IP packets between networks; a gateway also converts protocols.
3. Hub broadcasts to all ports; switch sends a frame only to the destination MAC port.

Long questions (board Section C)

1. Explain data communication channels with examples. Describe common networking devices.